MUFG INTERVIEW DRILLS

15 Q&As & Cheat Sheet

Technical Cross-Questions & Flashcards

Top 15 Interview Cross-Questions

Q1 [Algorithms]: What happens if there is a circular depe

WHY THE HIRING MANAGER ASKS THIS:
Tests understanding of DAG scheduling algorithms.

CRISP SENIOR MODEL ANSWER:
Kahn's algorithm tracks in-degrees. Cycle nodes never hit in-degree 0. The engine checks if len(ordered) != len(jobs) and raises ValueError immediately before executing any processes.

Q2 [Database]: Why use SQLite WAL mode instead of Pos

WHY THE HIRING MANAGER ASKS THIS:
Tests distributed systems & high availability knowledge.

CRISP SENIOR MODEL ANSWER:
For an on-host batch orchestrator agent, local storage prevents external network dependencies. WAL mode allows concurrent reads and writes without file-level locks.

Q3 [Reliability]: How do you handle a job that hangs indef

WHY THE HIRING MANAGER ASKS THIS:
Tests process lifecycle management under failure.

CRISP SENIOR MODEL ANSWER:
Pass timeout to subprocess.run(..., timeout=sec). On TimeoutExpired, catch exception, call proc.kill(), mark FAILED, and trigger an immediate P1 incident.

Q4 [SLA Governance]: Why measure wall-clock time inste

WHY THE HIRING MANAGER ASKS THIS:
Tests banking operational awareness.

CRISP SENIOR MODEL ANSWER:
Banking deadlines depend on absolute clock time. A job taking 0.5s of CPU but sleeping 2 minutes on retry delays the entire core banking window by 2 minutes.

Q5 [Testing]: How does job_02 handle attempt detection i

WHY THE HIRING MANAGER ASKS THIS:
Tests testing design and subprocess environment variables.

CRISP SENIOR MODEL ANSWER:
Inspects BATCH_ATTEMPT environment variable first, falling back to data/.job_02_recon_state.json. Attempt 1 fails with lock error; Attempt 2 succeeds and unlinks state.

Q6 [Unix OS]: Walk me through your df and free parsing c

WHY THE HIRING MANAGER ASKS THIS:
Tests practical Linux sysadmin competence.

CRISP SENIOR MODEL ANSWER:
We use awk to search for the token with % in df -h /, making it resilient to shifting columns.
In free -m, we check available memory (column 7) rather than raw free.

Q7 [Incident]: If /var/log hits 94% disk usage before EOD,

WHY THE HIRING MANAGER ASKS THIS:
Tests emergency L1 triage under pressure.

CRISP SENIOR MODEL ANSWER:
Run df -h to verify, du -ahx /var/log | sort -rh | head -20 to find runaway logs, lsof +L1 to check for deleted files held open by processes, and gzip old logs.

Q8 [Unix OS]: What are zombie processes and how do yo

WHY THE HIRING MANAGER ASKS THIS:
Tests OS kernel process lifecycle knowledge.

CRISP SENIOR MODEL ANSWER:
Zombies (state Z) have finished execution but their parent hasn't reaped them with wait(). Kill -9 fails on zombies. Send SIGCHLD to the parent or bounce the parent.

Q9 [Architecture]: Why check daemons like controlm-age

WHY THE HIRING MANAGER ASKS THIS:
Tests middleware awareness in financial networks.

CRISP SENIOR MODEL ANSWER:
Batch scripts rely on external bridges. Control-M agent receives instructions on port 7005; SWIFT MQ listener receives payments on IBM MQ channels. We verify them before batch start.

Q10 [Operations]: How do you prevent batch_engine.log f

WHY THE HIRING MANAGER ASKS THIS:
Tests logging best practices.

CRISP SENIOR MODEL ANSWER:
Configure RotatingFileHandler(maxBytes=50MB, backupCount=10) in Python, and configure Linux logrotate with daily, rotate 30, compress, and copytruncate.

Q11 [Banking]: Why didn’t you auto-retry JOB_04 when it

WHY THE HIRING MANAGER ASKS THIS:
Tests understanding of financial accounting integrity.

CRISP SENIOR MODEL ANSWER:
JOB_04 was deliberately blocked because EOD settlement failed. Retrying statutory reporting against uncommitted ledger balances causes invalid regulatory returns.

Q12 [Operations]: If JOB_03 is fixed at 3:15 AM, how do y

WHY THE HIRING MANAGER ASKS THIS:
Tests disaster recovery & operational restart procedures.

CRISP SENIOR MODEL ANSWER:
Query SQLite job_runs, confirm JOB_01 and JOB_02 are SUCCESS, bypass them, and resume directly from JOB_03, unblocking JOB_04 upon settlement completion.

Q13 [ITIL]: How does parent-child correlation stop Alert S

WHY THE HIRING MANAGER ASKS THIS:
Tests ITIL incident deduplication understanding.

CRISP SENIOR MODEL ANSWER:
Creates 1 primary P1 ticket for the root failure (JOB_03) and marks downstream blocked jobs (JOB_04) as child tickets on hold, focusing everyone on 1 bridge.

Q14 [ITIL]: Why raise a P2 ticket for JOB_02 if it ended wit

WHY THE HIRING MANAGER ASKS THIS:
Tests performance tuning vs functional success.

CRISP SENIOR MODEL ANSWER:
Functional success does not equal SLA compliance. Taking 3.20s vs 1.5s SLA window delays downstream cutoffs. P2 ticket alerts GL support to tune lock contention.

Q15 [Audit]: How does this report satisfy MAS and Fed au

WHY THE HIRING MANAGER ASKS THIS:
Tests regulatory and audit compliance knowledge.

CRISP SENIOR MODEL ANSWER:
MAS TRM and Fed FFIEC require immutable event logs and formal handovers. Our engine provides pre-batch OS telemetry, execution logs, tickets, and analyst sign-offs.

Rapid-Fire Flashcard Cheat Sheet

1. Jobs & Exit Codes

JOB_01_FX_INGEST: Exit 0 (Success, 0.63s, SLA 5.0s)

JOB_02_LEDGER_RECON: Exit 0 (SLA BREACH 3.20s > 1.5s)

JOB_03_EOD_SETTLEMENT: Exit 2 (FAILED, Deadlock ORA-00060)

JOB_04_REG_REPORT: Exit - (BLOCKED_BY_UPSTREAM)

2. Incidents & Runbooks

INC0010404 (P1): JOB_03 | Core-Banking-DBA-L3 | KB0041209

INC0010405 (P2): JOB_02 | FinControl-GL-Support-L2 | KB0038912

INC0010406 (P3): JOB_04 | Regulatory-Reporting-L2 | Parent: INC00104

3. Linux Telemetry & Daemons

Disk Usage: df -h / (< 85% threshold, 0.0% measured)

Memory Usage: free -m (< 90% threshold, 70.8% measured)

CPU Load (1m): /proc/loadavg (< 4.0 threshold, 0.96 measured)

Zombies: ps aux | awk '$8=="Z"' (0 detected)

Spool Purge: 3 files purged from data/spool_tmp/

Daemons Verified: controlm (7005), swift-mq, as400

4. CLI Commands Quick Reference

Run Full Batch: python3 main.py --run-batch

Inspect Database: python3 main.py --show-db

View CSV Report: cat reports/Daily_Ops_CheckSheet_*.csv

View HTML Report: google-chrome reports/Shift_Handover_*.html

View Outbox REST: cat data/servicenow_outbox/INC0010404.json